



COMP47350: Data Analytics (Conv)

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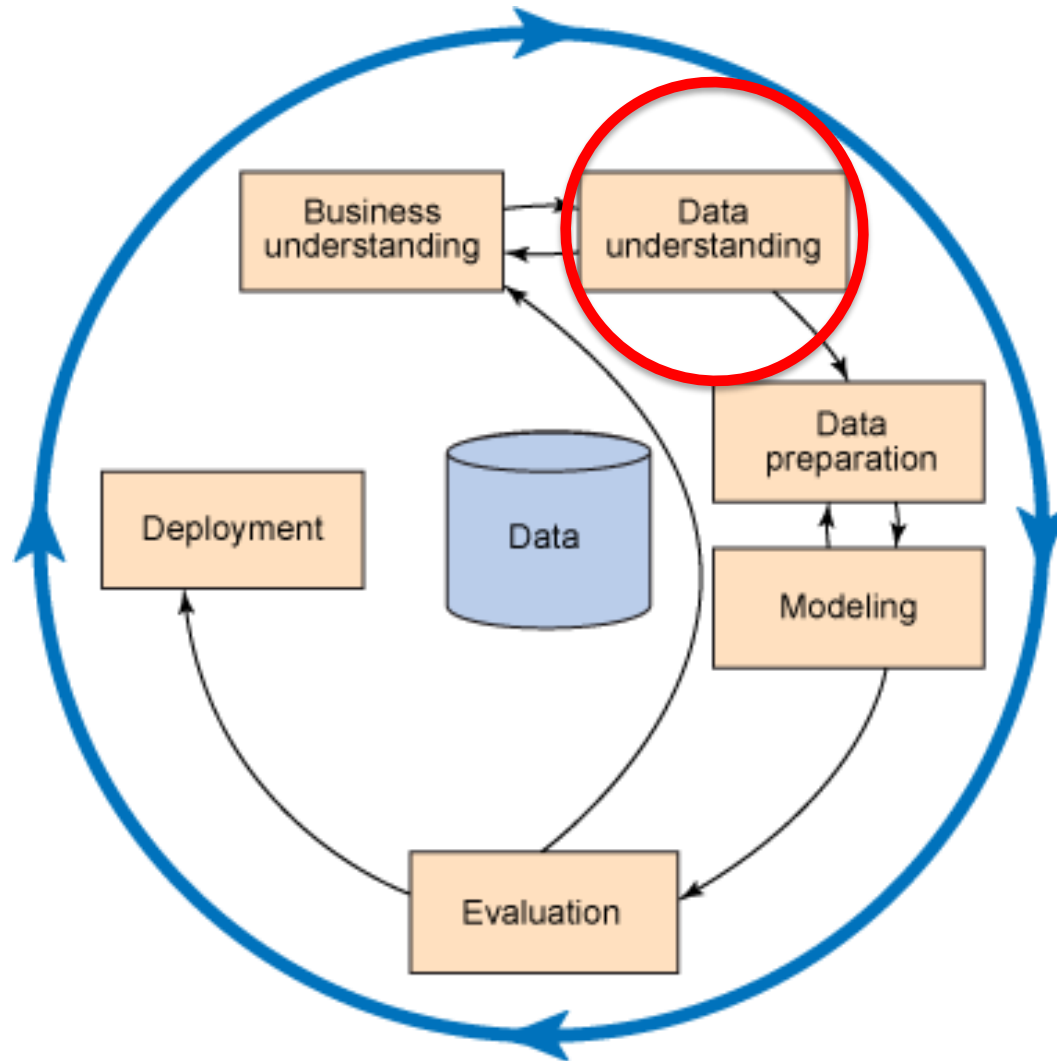
Module Topics

- **Python Environment** (Anaconda, Jupyter Notebook)
- **Getting Data** (Web scrapping, APIs, DBs)
- **Understanding Data** (stats, visualisation)
- **Preparing Data** (cleaning, transformation)
- **Modeling & Evaluation** (machine learning algorithms)

Data Analytics Project Lifecycle:

CRISP-DM

CRISP-DM: **C**Ross-**I**ndustry **S**tandard **P**rocess for **D**ata **M**ining



Case Study



Motor Insurance Claim Fraud Detection

What is the business problem?

Despite having a fraud investigation team that investigates up to 30% of all claims, **a motor insurance company is still losing money due to fraudulent claims.**

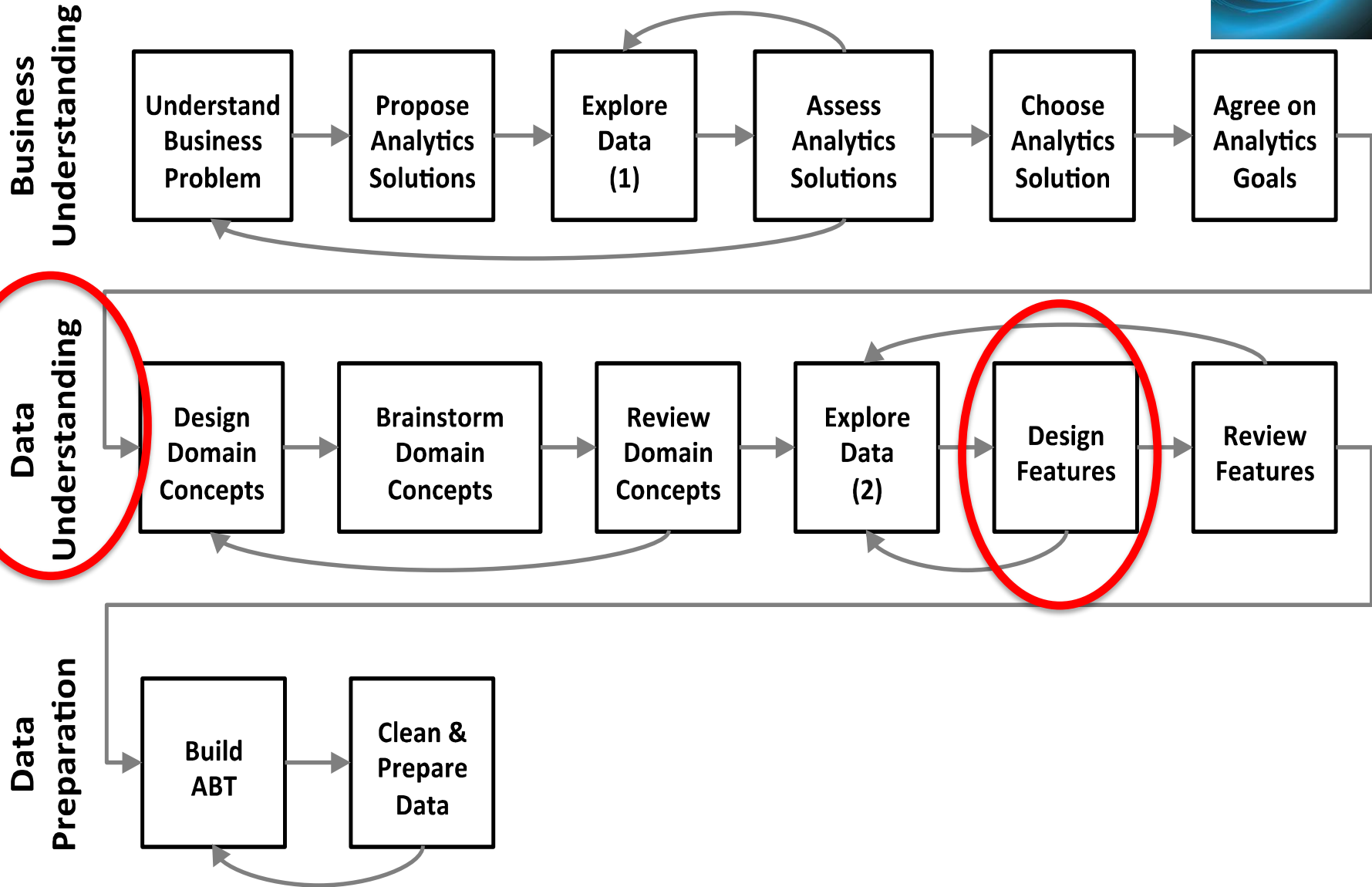
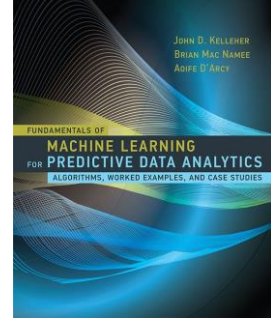
Analytics solution:

Claim Prediction: Given a specific **claim**, is it likely to be fraudulent or not?

Input (descriptive features): claim and claimant description (aka historical data)

Output (target feature): **fraud/not-fraud label** (aka human supervision)

CRISP-DM Summary Diagram: First 3 Stages (see textbook)



CRISP-DM: Data Understanding

Designing the Dataset: CSV file describing training examples via descriptive features and the target feature; **1 training example per row**

Designing Features:

- **Different Data Types** (e.g., numeric, text, ordinal): **we group features into two main types**
 - **continuous features:** can compute summary statistics such as min, max, average. **Example: price** (numeric features).
 - **categorical features:** can compute summary stats such as counts, e.g., most frequent category, **Example: country name, age group** (ordinal, binary, categorical, textual)
- **Different Feature Types** (e.g., raw features or derived features)

Designing Features

- **Feature** = any measure derived from a **domain concept**
 - concept: **Demographics**, feature: **Age**
- **Key considerations when designing features:**
 - **Data availability:** is data for this feature available? (e.g., does the company collect **Age** info?)
 - **Timing:** when does the data for this feature become available? (e.g., is feature **Age** available when the Claim is made?)
 - **Longevity:** how quickly will this feature go stale? (e.g., for predicting outcomes of loans as repaid/default, feature **Income** changes over the years, but the ratio of **Income/Loan** is more stable)
 - **Usefulness:** does this feature capture some useful information about the target feature? (e.g., features that are useful for predicting the target: e.g., feature **Name** not so useful, **Income** could be more useful)
- DISCUSSION

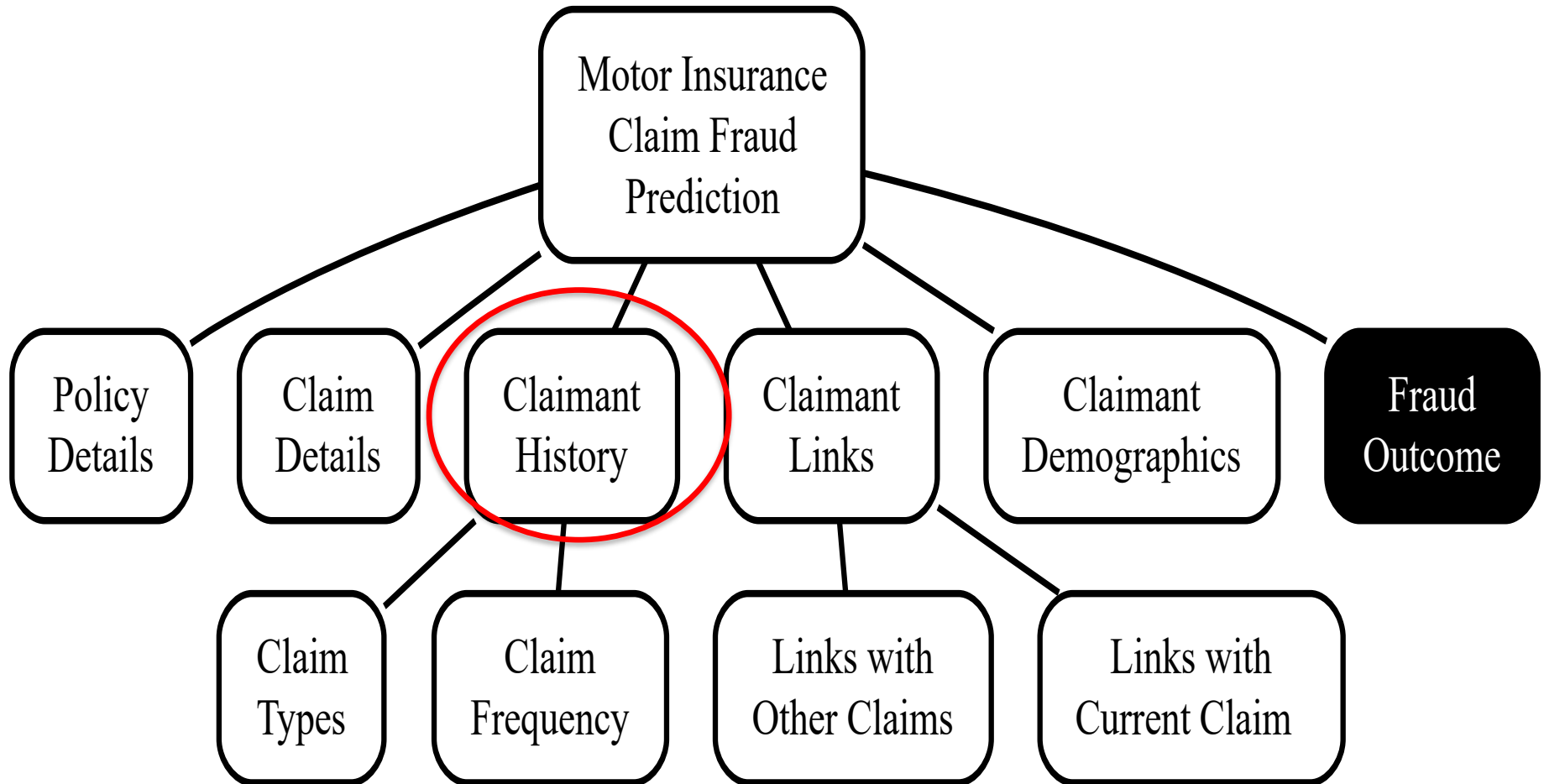
Designing Features

- Features can be of two types:
 - **Raw features** (e.g., exact salary)
 - **Derived features (engineered features)**
 - **Aggregates** (e.g., average salary, salary range)
 - **Flags** (yes/no or 0/1 flag for: was bank account overdrawn?)
 - **Ratios** (salary-to-loan ratio)
 - **Mappings** (e.g., map exact salary to category: high, medium, low salary)

Handling Time

- **Time/season** may influence behavior, so time-dependent derived features should be considered
- **Recency** of data could be important (e.g., a feature for the claim history over past 12 months or past 6 months or by quarters)

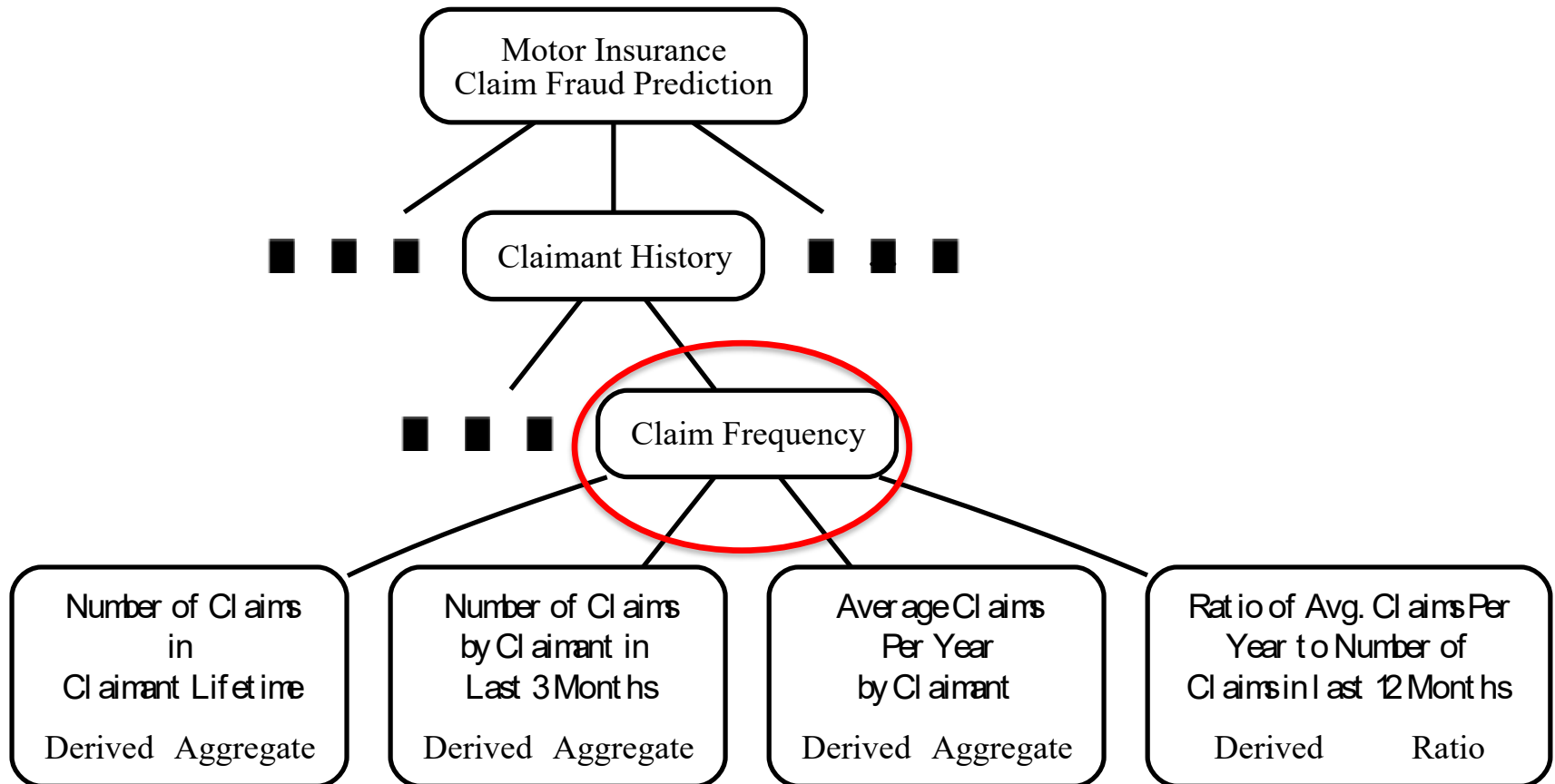
Implementing Features



Try to capture past behavior of claimant and relation to target feature

Implementing Features

Example features that describe the **Claimant History** domain concept



Data Understanding

Main Tools for Data Understanding:

- Data Quality Report (DQR)
- Data Quality Plan (DQP)

Data Understanding

- **Data Quality Report**

- Getting To Know The Data (analyse each feature)
- Identifying Data Quality Issues (e.g., data integrity checks, problems affecting each feature)
- An initial summary of the dataset, with discussion of issues identified (**written report documenting all the issues**)

- **Data Quality Plan**

- Handling Data Quality Issues (how to deal with missing values, outliers, errors in the data)
- A record of problems and solutions for each feature (**action and implementation plan; record of how data was cleaned**)

Data Quality Report

- **Tables** (e.g., descriptive stats):
 - Describe the features in the dataset, using standard **statistical measures** of central tendency and variation
- **Visualizations** (e.g., data plots):
 - Individual features: A **histogram**, **boxplot** or **barplot** for each feature in the dataset (plot depends on feature type)
 - Pairs of features: **scatter plot**, **barplot**: we look for feature correlation

Data Quality Report

Tables:

- **Step1: Group the features into continuous or categorical types** (done based on data dictionary describing each feature and further understanding of the data)
- **Step2: Prepare two summary tables**, one for continuous features one for categorical features
 - Compute different stats for **continuous features** (e.g., average Salary) versus **categorical features** (e.g., most frequent Country)

Data Quality Report

Tables:

- For continuous features we compute:
 - **Central tendency** (e.g., mean, median, mode)
 - **Variation** (e.g., min-max range, variance, percentiles)
- For categorical features we compute:
 - **Counts** (e.g., most popular category, unique values for each category)

Data Quality Report: Tables

Continuous features:

Feature: Feature name

Count: Number of rows with this feature

%Miss: Percentage of missing values

Card: Number of unique values

Min: min value for this feature

1st qrt: 25th percentile

Mean: average value

Median: middle value or 50th percentile

3rd qrt: 75th percentile

Max: max value

Std Dev: standard deviation

(a) Continuous Features

[illegible]

Motor Insurance Fraud

500 rows: **1 example per row** (claim details and FraudFlag outcome)

14 columns: **1 feature per column** (13 descriptive features, 1 target)

ID	TYPE	INC	MARITAL STATUS	NUM CLMNTS	INJURY TYPE	HOSPITAL STAY	CLAIM AMNT	TOTAL CLAIMED	NUM CLAIMS	NUM SOFT TISS	% SOFT TISS	CLAIM AMT RCVD	FRAUD FLAG
1	CI	0		2	Soft Tissue	No	1,625	3250	2	2	1.0	0	1
2	CI	0		2	Back	Yes	15,028	80,112	1	0	0	15,028	0
3	CI	54,613	Married	1	Broken Limb	No	-99,999	0	0	0	0	572	0
4	CI	0		4	Broken Limb	Yes	5,097	11,661	1	1	1.0	7,864	0
5	CI	0		4	Soft Tissue	No	8869	0	0	0	0	0	1
6	CI	0		1	Broken Limb	Yes	17,480	0	0	0	0	17,480	0
7	CI	52,567	Single	3	Broken Limb	No	3,017	18,102	2	1	0.5	0	1
8	CI	0		2	Back	Yes	7463	0	0	0	0	7,463	0
9	CI	0		1	Soft Tissue	No	2,067	0	0	0	0	2,067	0
10	CI	42,300	Married	4	Back	No	2,260	0	0	0	0	2,260	0
300	CI	0		2	Broken Limb	No	2,244	0	0	0	0	2,244	0
301	CI	0		1	Broken Limb	No	1,627	92,283	3	0	0	1,627	0
302	CI	0		3	Serious	Yes	270,200	0	0	0	0	270,200	0
303	CI	0		1	Soft Tissue	No	7,668	92,806	3	0	0	7,668	0
304	CI	46,365	Married	1	Back	No	3,217	0	0	0	0	1,653	0
458	CI	48,176	Married	3	Soft Tissue	Yes	4,653	8,203	1	0	0	4,653	0
459	CI	0		1	Soft Tissue	Yes	881	51,245	3	0	0	0	1
460	CI	0		3	Back	No	8,688	729,792	56	5	0.08	8,688	0
461	CI	47,371	Divorced	1	Broken Limb	Yes	3,194	11,668	1	0	0	3,194	0
462	CI	0		1	Soft Tissue	No	6,821	0	0	0	0	0	1
491	CI	40,204	Single	1	Back	No	75,748	11,116	1	0	0	0	1
492	CI	0		1	Broken Limb	No	6,172	6,041	1	0	0	6,172	0
493	CI	0		1	Soft Tissue	Yes	2,569	20,055	1	0	0	2,569	0
494	CI	31,951	Married	1	Broken Limb	No	5,227	22,095	1	0	0	5,227	0
495	CI	0		2	Back	No	3,813	9,882	3	0	0	0	1
496	CI	0		1	Soft Tissue	No	2,118	0	0	0	0	0	1
497	CI	29,280	Married	4	Broken Limb	Yes	3,199	0	0	0	0	0	1
498	CI	0		1	Broken Limb	Yes	32,469	0	0	0	0	16,763	0
499	CI	46,683	Married	1	Broken Limb	No	179,448	0	0	0	0	179,448	0
500	CI	0		1	Broken Limb	No	8,259	0	0	0	0	0	1

Data Quality Report: Tables

Motor Insurance Claim Prediction

(a) Continuous Features

Feature	Count	%			1 st			3 rd		Std.
		Miss.	Card.	Min	Qrt.	Mean	Median	Qrt.	Max	Dev.
INCOME	500	0.0	171	0.0	0.0	13,740.0	0.0	33,918.5	71,284.0	20,081.5
NUM CLAIMANTS	500	0.0	4	1.0	1.0	1.9	2	3.0	4.0	1.0
CLAIM AMOUNT	500	0.0	493	-99,999	3,322.3	16,373.2	5,663.0	12,245.5	270,200.0	29,426.3
TOTAL CLAIMED	500	0.0	235	0.0	0.0	9,597.2	0.0	11,282.8	729,792.0	35,655.7
NUM CLAIMS	500	0.0	7	0.0	0.0	0.8	0.0	1.0	56.0	2.7
NUM SOFT TISSUE	500	2.0	6	0.0	0.0	0.2	0.0	0.0	5.0	0.6
% SOFT TISSUE	500	0.0	9	0.0	0.0	0.2	0.0	0.0	2.0	0.4
AMOUNT RECEIVED	500	0.0	329	0.0	0.0	13,051.9	3,253.5	8,191.8	295,303.0	30,547.2
FRAUD FLAG	500	0.0	2	0.0	0.0	0.3	0.0	1.0	1.0	0.5

Look at each feature one by one.

Can you point out some issues affecting the features in this summary?

Data Quality Report: Tables

Categorical features:

Mode: most frequent value

Mode Freq: frequency of most frequent value

Mode%: percentage of most frequent value

2nd Mode: second most frequent value

(b) Categorical Features

[illegible]

Data Quality Report: Tables

Motor Insurance Claim Prediction

(a) Categorical Features

Feature	Count	%		Mode	Mode Freq.	Mode %	2 nd	2 nd
		Miss.	Card.				Mode Freq.	Mode %
INSURANCE TYPE	500	0.0	1	CI	500	1.0	–	–
MARITAL STATUS	500	61.2	4	Married	99	51.0	Single	48 24.7
INJURY TYPE	500	0.0	4	Broken Limb	177	35.4	Soft Tissue	172 34.4
HOSPITAL STAY	500	0.0	2	No	354	70.8	Yes	146 29.2

Can you point out some issues affecting the features in this summary?

Why do we need summary tables?

References

- **FMLPDA Book: Fundamentals of Machine Learning for Predictive Data Analytics**, John D. Kelleher, Brian Mac Namee, Aoife D'Arcy, MIT Press, 2015 (<http://machinelearningbook.com/>)